

Seeing around the corner before the car can

Roadside Assistant for hidden pedestrians and cyclists in CARLA



Occlusion turns a visible road user into a late emergency

What the car misses

Parked vehicles can hide the last metres of a crossing from the approaching car.

What the roadside camera adds

The roadside camera sees from an elevated angle and shares that view before the vehicle's own camera can react.

The system adds an earlier layer of perception

01

See

An RSU camera detects pedestrians and cyclists behind the occluder.

02

Share

Compact, time-stamped object tracks travel over MQTT.

03

Act

The car checks whether the road user will enter its path, then slows or brakes.

One loop connects infrastructure, communication, and control

01 RSU detects, projects to world coordinates, tracks, and publishes.

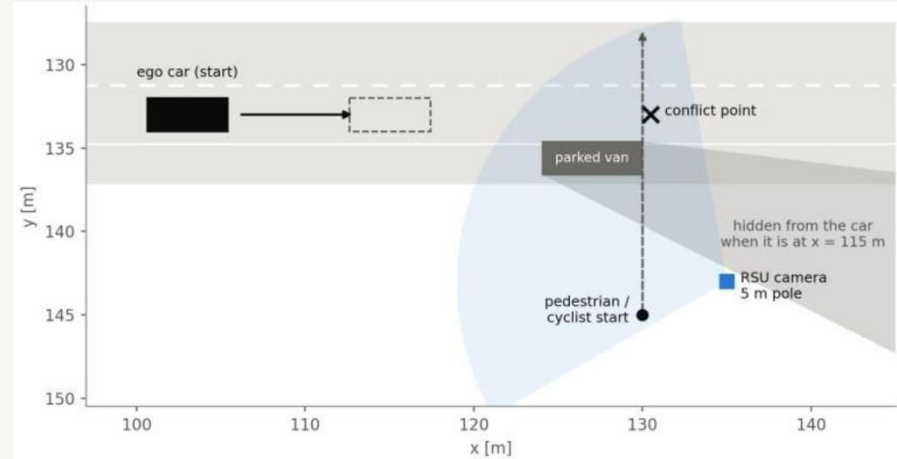
02 The link delivers the warning after an emulated 100 ms radio delay.

03 The vehicle advances the object to the current time, checks path overlap, and applies the policy.

04 CARLA's synchronous clock keeps the experiment repeatable at 25 Hz.

The elevated camera is positioned for the hidden zone

- Town03 scenario: the car approaches a crossing hidden by a parked van or truck.
- The RSU sits on a 5 m pole with a 110° field of view, covering the crossing, both lanes, and the occluded area.
- The car's view misses the road user until it is already near the lane; the RSU sees the approach earlier.



From pixels to a warning the car can use

01

Detect

YOLOv8n keeps person and bicycle detections and merges a rider with a bicycle into one cyclist.

02

Position

The bottom centre of each box is back-projected onto the road plane to estimate world position.

03

Track

A nearest-neighbour tracker adds stable IDs, velocity, prediction, and short-term coasting.

04

Publish

The message is about 123 bytes for one object and includes position, velocity, confidence, and capture time.

The car reacts only when the warning intersects its path

SHARED CHECK

Same path check

The same path check evaluates RSU and car-camera objects.

Shared timestamps support delay, age compensation, and freshness checks.

CAUTION

RSU-only threat

Slow to 15 km/h while the road user is still hidden.

DANGER

The car camera confirms

Brake to stop 2 m before a relevant road user.

Warned about pedestrians it cannot see

Car camera (assisted, S3)



RSU camera at the same moment



The orange circles are RSU positions projected into the car's image: the pedestrians are still behind the van. The banner shows the source (RSU) and the age of the information (160 ms).

Three deliberately difficult crossings test the difference

S1

Pedestrian · van

One pedestrian behind a parked van, clear noon.

S2

Cyclist · truck

One cyclist behind a parked truck, clear noon.

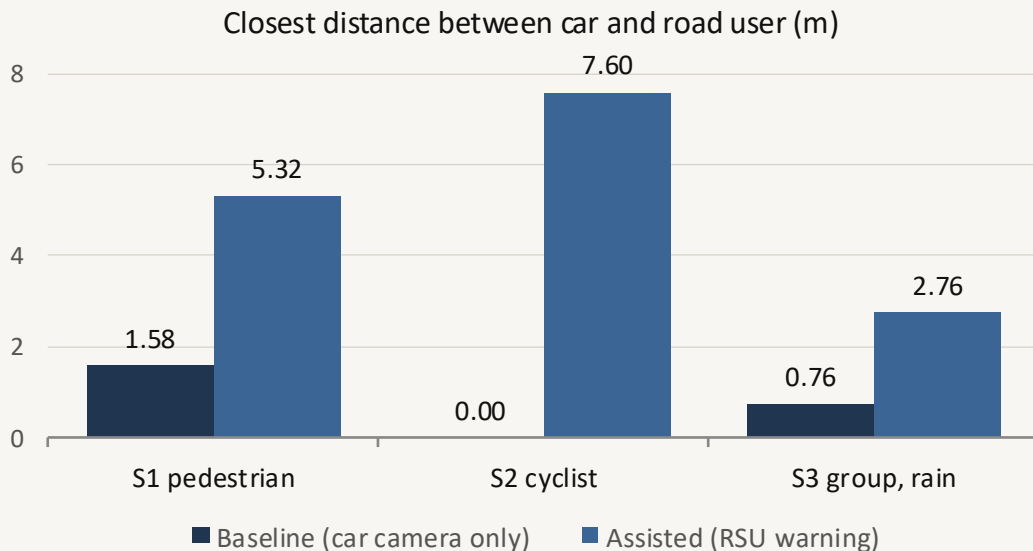
S3

Group · rain

Fifteen pedestrians behind a van in heavy rain at sunset.

Each scenario runs twice with an identical setup: baseline (car camera only) versus assisted (RSU warning).

The RSU turned every tested conflict into a safe outcome



Outcome: baseline → assisted

S1 near miss → safe

S2 collision → safe

S3 near miss → safe

A near miss is a closest distance under 2 m without contact. The bars show how close the road user came

Without assistance, the car never braked in S1 or S3: its camera did not see the road user in time, and they passed 1.58 m and 0.76 m from the moving car. In S2 the camera saw the cyclist only 3.3 m before the crossing, too late to avoid the collision.

Same crossing, with and without the warning

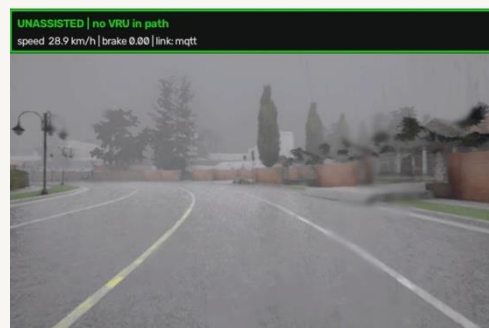
S1 pedestrian



S2 cyclist



S3 group, rain



Baseline

car camera only



Assisted

with RSU
warning

The warning arrived while there was still room to act

1.24 s **2.41 s** **2.16 s**

S1 pedestrian

S2 cyclist

S3 group, rain

RSU warning before the car would reach the crossing (baseline runs)

That equals roughly 9.7–19.4 m of road ahead.

Where the car camera saw the road user at all (S2), the RSU warned 2.0 s earlier: a 2.0 s lead time over the car's own camera.

